

Question ID: f6cbb04a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

$d = 55t$

The equation above can be used to calculate the distance d , in miles, traveled by a car moving at a speed of 55 miles per hour over a period of t hours. For any positive constant k , the distance the car would have traveled after $9k$ hours is how many times the distance the car would have traveled after $3k$ hours?

Answer

- A. 3
- B. 6
- C. $3k$
- D. $6k$

Correct Answer: A

Rationale

Choice A is correct. Since the distance is equal to the amount of time multiplied by a constant, the given equation $d = 55t$ represents a proportional relationship between distance and time in this situation. Since $9k = 3 \cdot 3k$, the time when $t = 9k$ hours is 3 times the time when $t = 3k$ hours. Therefore, the distance traveled after $9k$ hours is 3 times the distance after $3k$ hours.

Choices B and D are incorrect and may result from interpreting the proportional relationship between time and distance as additive rather than multiplicative. Choice C is incorrect and may result from an arithmetic error.